Multimedia Systems And Applications

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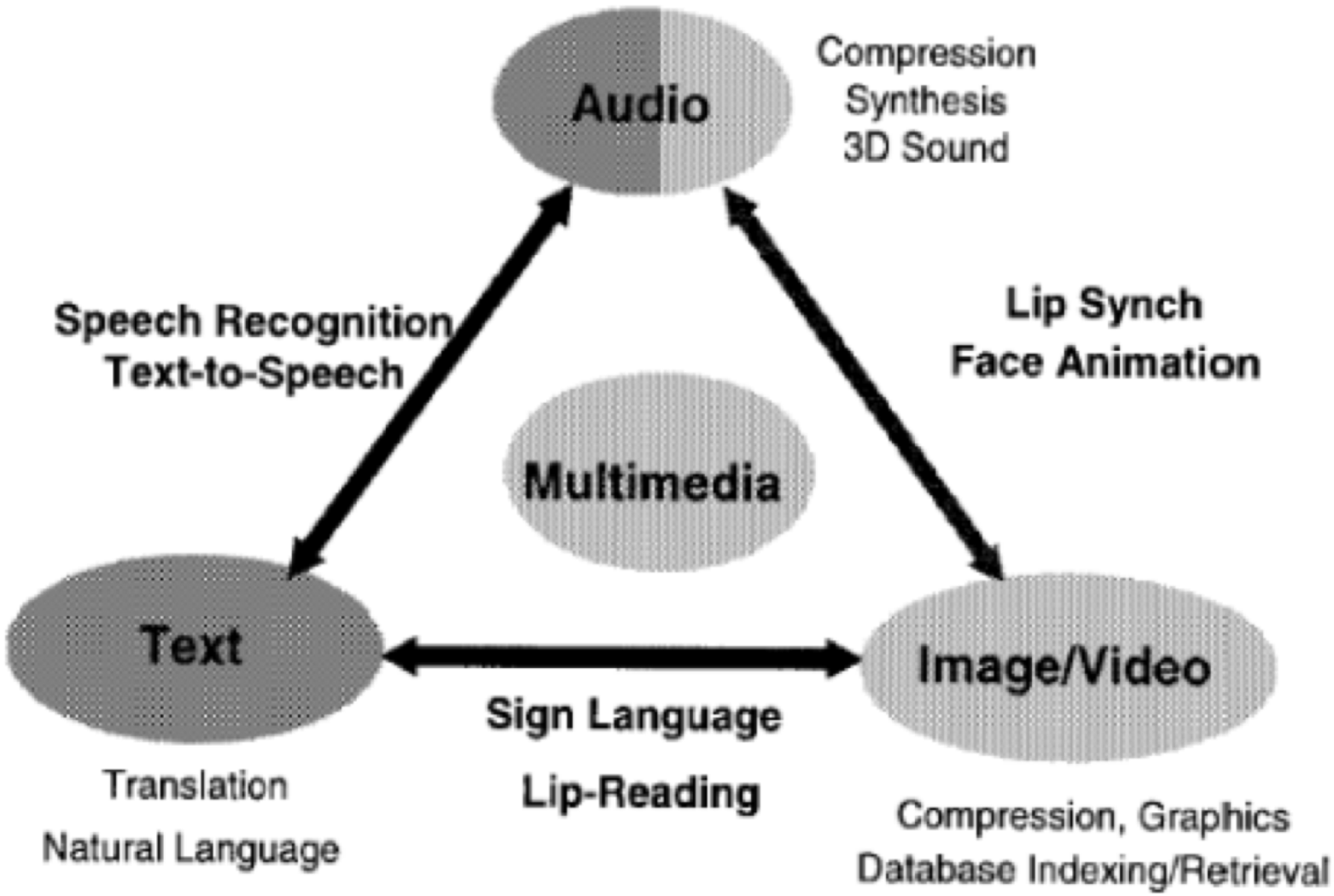
Introduction Multimedia

- A **medium** refers to different type of data representation such as text, images, graphics, speech, audio, video.
- The term “**Multimedia**” simply means "more than one medium." In other words television programs, movies, even illustrated books are all examples of multimedia.
- Multimedia uses a combinations of text, images, sounds, and movement.
- Multimedia is a mixture of different forms of media. This includes text, graphics, audio, video, etc.

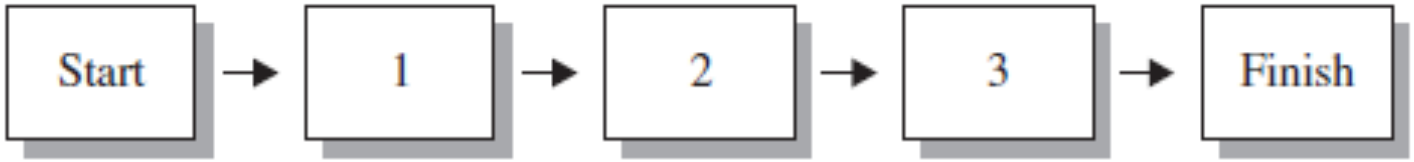
Interaction of Media in Communication



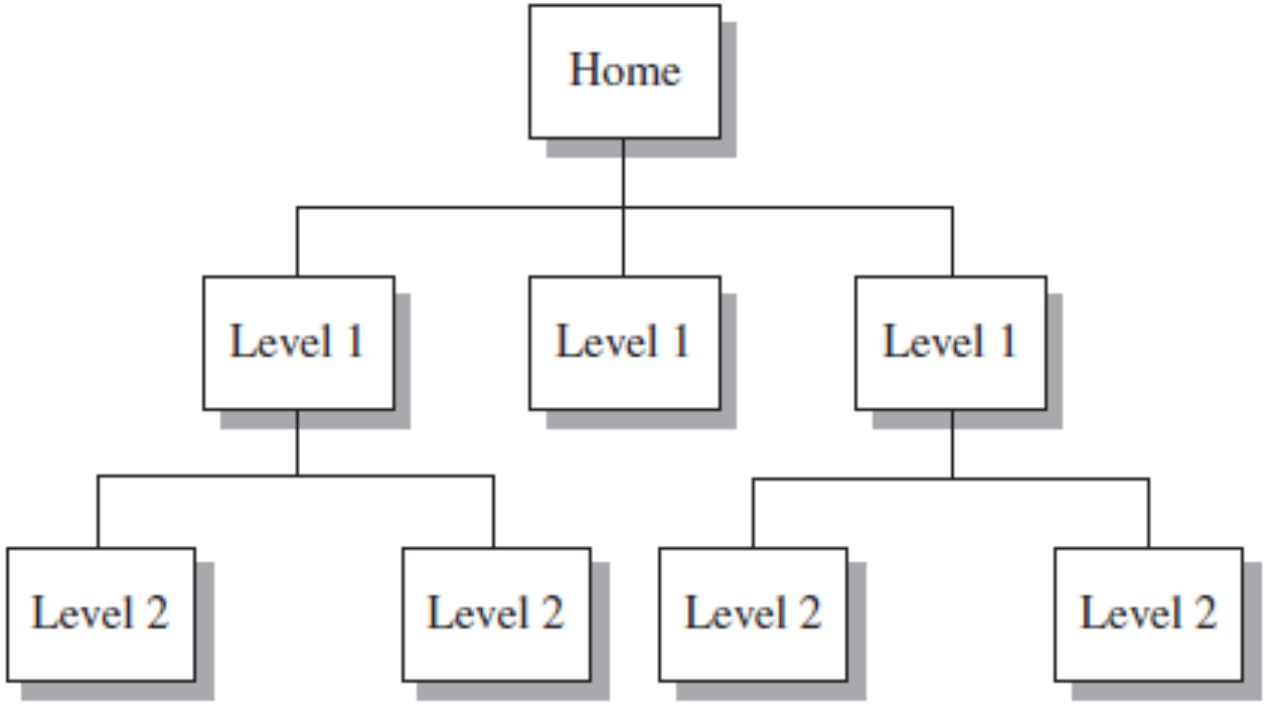
Categories of Multimedia

- Based on how multimedia programs are used, multimedia can be divided into two forms – **linear multimedia** and **non-linear** multimedia.
- In **linear multimedia**, information is read or viewed in a continuous sequence.
- Usually, these presentations begin at a predetermined starting point and end at a predetermined end point. They can be automated so that each screen comes after a fixed time interval.
- Example: Powerpoint presentation is one of the most common examples of linear multimedia.
- On the other hand, **non-linear multimedia** information is not presented in sequential or chronological manner. Non-linear multimedia programs are usually interactive and require audience interaction

Linear Multimedia



Nonlinear Multimedia



Features of Multimedia

i. Interactivity
ii. Synchronization
iii. Hyperlinking
iv. Scalability
v. Compression
vi. Rich Visual Effects
vii. Simulations and Virtual Environments
viii. Streaming and On-Demand Access
ix. Personalization

Applications of Multimedia

- Business
- Research and Medicine
- Public Access Notes
- Entertainment
- Industry
- Commercial
- Education
- Engineering

Stages of Multimedia Project

1. **Pre-production**: The process before producing multimedia project.
2. **Production**: The process in which multimedia project is produced.
3. **Post-production**: The process after the production of multimedia project

Pre-production

- Idea or Motivation
- Product Concept and Project Goals
- Target Audience
- Delivery Medium and Authoring Tools
- Planning-
    - What do you require for the multimedia project?
    - How long will each task take?
    - Who is going to do the work?
    - How much will the product cost?

Production

- Scriptwriting
- Art
- 3D Modelling and Animation
- Authoring
- Shooting and Recording Digitizing Video
- Quality Control

Post-production

- Testing
- Mastering
- Archiving and Duplication
- Marketing and Distribution

Multimedia Skills Required in a Team

- Project manager
- Multimedia designer
- Interface designer
- Multimedia programmer
- Computer programmers
- Writer Subject matter expert
- Audio specialist
- Video specialist
- Producer for the Web
- Permission specialist

Skills Required for Various Team Members

The project manager is responsible for:

- The overall development, implementation, and day-to-day operations of the project.
- The design and management of a project.
- Understanding the strengths and limitations of hardware and software.
- Make schedules.
- Decide the budget of the project.
- Interact with team and clients.
- Provides resolution to development and production problems.
- Motivate people and should be detail oriented.

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Skills Required for Various Team Members

An interface designer is responsible for:

- Creating a software device that organizes content.
- It allows users to access or modify content, and presents that content on the screen.
- Building a user-friendly interface

Skills Required for Various Team Members

A multimedia writer is responsible for:

- Notes Creating characters, actions, point of view, and interactivity.
- Writing proposals and test screens.
- Scripting voice-overs and actors’ narrations.

A video specialist needs to understand:

- The delivery of video files on CD, DVD, or the Web.
- How to shoot quality video.
- How to transfer the video footage to a computer.
- How to edit the footage down to a final product using digital non-linear editing system (NLE).

Skills Required for Various Team Members

An audio specialist is responsible for:

- Locating and selecting suitable music talent.
- Scheduling recording sessions.
- Digitizing and editing recorded material into computer files

Skills Required for Various Team Members

A multimedia programmer is responsible for:

- Locating audio/video resources.
- Selecting suitable audio/video clips.
- Creating audio/video clips.
- Interacting with project managers and instructional designers.
- Participating in the design process.
- Working on storyboard and uses it as a guideline.
- Finding out problems, solving them and fixing bugs.
- Writing understandable, easy and reusable codes
- Liaising with designers
- Integrates all the multimedia elements into a seamless project, using authoring systems or programming language.
- Manages timings, transitions and record keeping.

Basic tools supporting Multimedia development include:

- **World Wide Web (browser-based playback)** - HTML 5, CSS, Flash, Dreamweaver
- **Programming Applications (native playback)** - Director, LiveCode
- **Artwork Development** - Photoshop, Illustrator
- **Animation Development (2D and 3D)** - After Effects, Maya
- **Audio Editing** - Sound Forge, Audition
- **Video Editing** - Final Cut Pro, Premiere Pro
- **Compression** - Adobe Media Encoder, Sorenson Squeeze

Identifying Multimedia elements

- There are five basic elements of multimedia: **text, images, audio, video and animation**.
- Example - Text in fax, Photographic images, Geographic information system maps, Voice commands, Audio messages, Music, Graphics, Moving graphics animation, Full-motion stored and live video, Holographic images.
- Text and images are static objects, whereas audio, video and animations are dynamic objects that move or change.

Text

- Text is most commonly used to communicate information.
- It has alphanumeric characters, in addition to special characters.
- It involves the use of text types, sizes, fonts, colours and background colours.
- Multimedia applications support linked content, through Hypertext.
- Text in SMS, FAX, Email are examples of this element in Communication.
- Common file types include: TXT, DOC, DOCX, PDF.

Images

- Images/Illustrations are the oldest form of media
- They help to illustrate ideas through still pictures.
- There are two types of images - Bitmaps and Vector
- Bitmap images are real images that can be captured from devices such as cameras.
- Vector graphics are created using software in the computer .
- This multimedia element enables to generate, represent, process, manipulate, and display pictures.
- Common file types for Images include: JPG, PNG, TIF, BMP

Audio

- The speech, music and sound effects used in multimedia is digital audio.
- Multimedia applications use audio or the sound element like, website or presentation can add audio files from a musical background, or a voiceover / spoken explanation.
- There are two basic type of audio or sound; analog and digital.
- The original sound signal is termed as Analog audio.
- The digital sampling of the original sound is termed as Digital audio.
- Common file types for Audio include: MP3, WAV, WMA

Video

- Video presents moving pictures and typically combines images and sound for a multimedia experience.
- This technology records, synthesizes, and displays images known as frames in such sequences at a fixed speed that makes the creation appear as moving; this is how we see a completely developed video.
- To watch a video without any interruption, video device must display 25 to 30 frames/second.
- Common file types for Video include AVI, WMV, FLV, MOV, MP4

Animation

- Animation is the process of making a static image to look as if it is moving.
- It helps in creating, developing, sequencing, and displaying a set of images technically known as frames.
- Digital animation can be classified as 2D two dimension and 3D Three dimension animations.
- GIFs, an abbreviation for graphic image files, are small files that present a single image or rapidly display a sequence of a few images to give the appearance of motion.
- AdobeFlash is the most common tool for creating these animations.
- Common file types for Animation include: GIF, FLV

Multimedia Hardware

- Most of the computers now-a-days come equipped with the hardware components required to develop/view multimedia applications.
- Following are the various categories in which we can define the various types of hardwares required for multimedia applications.

Processor

- The heart of any multimedia computer is its processor.
- Today Core 15 or higher processor is recommended for a multimedia computer.
- CPU is considered as the brain of the computer.
- CPU performs all types of data processing operations.
- It stores data, intermediate result and instructions (program).
- It controls the operations of all parts of computer.

Multimedia Hardware

- **Memory and Storage Devices** - You need memory for storing various files used during production, original audio and video clips, edited pieces and final mined pieces. You also need memory for backup of your project files.
  1. **Primary Memory**- Primary memory holds only those data and instructions on which computer is currently working. It has limited capacity and data gets lost when power is switched off. It is generally made up of semiconductor device. These memories are not as fast as registers. The data and instructions required to be processed earlier reside in main memory. It is divided into two subcategories RAM and ROM.

Multimedia Hardware: Memory and Storage Devices

2. **Flash Memory**- Cache memory is a very high speed semiconductor memory, which can speed up CPU. It acts as a buffer between the CPU and main memory. It is used to hold those parts of data and program which are most frequently used by CPU. The parts of data and programs are transferred from disk to cache memory by operating system, from where CPU can access them.
3. **Secondary Memory:** This type of memory is also known as external memory or non-volatile. It is slower than main memory. These are used for storing Data/Information permanently. CPU directly does not access these memories; instead they are accessed via input-output routines. Contents of secondary memories are first transferred to main memory and then CPU can access it. For example, disk, CD-ROM, DVD, etc.

Multimedia Hardware

- **Input Devices** - Following are the various types of input devices which are used in multimedia systems.
  1. **Keyboard**
  2. **Mouse**
  3. **Joystick**
  4. **Light Pen**
  5. **Track Ball**
  6. **Scanner**
  7. **Voice Systems**
  8. **Digital Video Camera**

Multimedia Hardware

- **Output Devices**
  1. **Monitors**
  2. **Printers**
  3. **Screen Projectors**
  4. **Speakers and Sound Card**

Multimedia Software

- Multimedia software tells the hardware what to do. For example, multimedia software tells the hardware to display the color blue, play the sound of cymbals crashing etc.
- To produce these media elements(movies, sound, text, animation, graphics etc.) there are various software available in the market such as Paint Brush, Photo Finish, Animator, Photo Shop, 3D Studio, Corel Draw, Sound Blaster, IMAGINET, Apple Hyper Card, Photo Magic, Picture Publisher.

The categories of basic software tools are:

Text Editing Tools- These tools are used to create letters, resumes, invoices, purchase orders, user manual for a project and other documents. MS-Word is a good example of text tool. It has following features:

1. Creating new file, opening existing file, saving file and printing it.
2. Insert symbol, formula and equation in the file.
3. Correct spelling mistakes and grammatical errors.
4. Align text within margins.
5. Insert page numbers on the top or bottom of the page.
6. Mail-merge the document and making letters and envolpes.
7. Making tables with variable number of columns and rows.

The categories of basic software tools are:

Painting and Drawing Tools- These tools generally come with a graphical user interface with pull down menus for quick selection. You can create almost all kinds of possible shapes and resize them using these tools.

Drawing file can be imported or exported in many image formats like .gif, .tif, .jpg, .bmp, etc. Some examples of drawing software are Corel Draw, Freehand, Designer, Photoshop, Fireworks, Point etc.

These software have following features:

1. Tools to draw a straight line, rectangular area, circle etc.
2. Different colour selection option.
3. Pencil tool to draw a shape freehand.
4. Eraser tool to erase part of the image.
5. Zooming for magnified pixel editing.

The categories of basic software tools are:

Image Editing Tools- Image editing tools are used to edit or reshape the existing images and pictures.

- These tools can be used to create an image from scratch as well as images from scanners, digital cameras, clipart files or original artwork files created with painting and drawing tools.
- Examples of Image editing or processing software are Adobe Photoshop and Paint Shop Pro.

The categories of basic software tools are:

Sound Editing Tools- These tools are used to integrate sound into multimedia project very easily. You can cut, copy, paste and edit segments of a sound file by using these tools.

- The presence of sound greatly enhances the effect of a mostly graphic presentation, especially in a video.
- Examples of sound editing software tools are: Cool Edit Pro, Sound Forge and Pro Tools.
- These software have following features:
  1. Record your own music, voice or any other audio.
  2. Record sound from CD, DVD, Radio or any other sound player.
  3. You can edit, mix the sound with any other audio.
  4. Apply special effects such as fade, equalizer, echo, reverse and more.

The categories of basic software tools are:

Video Editing Tools- These tools are used to edit, cut, copy, and paste your video and audio files.

- Video editing used to require expensive, specialized equipment and a great deal of knowledge.
- The aritistic process of video editing consists of deciding what elements to retain, delete or combine from various sources so that they come together in an organized, logical and visually planning manner.
- Today computers are powerful enough to handle this job, disk space is cheap and storing and distributing your finished work on DVD is very easy. Examples of video editing software are Adobe Premiere and Adobe After Effects.